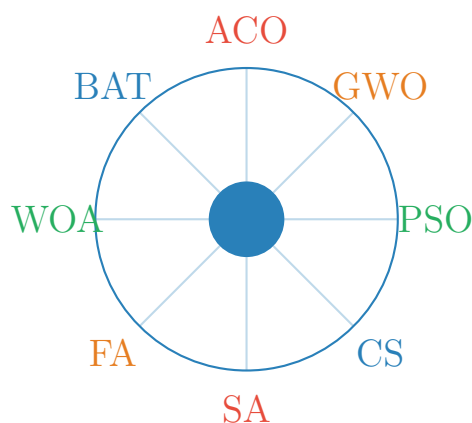


Metaheuristic Optimization for Deep Learning & Explainable AI



Sentiment Analysis on Hotel Reviews

Using TripAdvisor Dataset with 7,000+ Samples

Algorithms Implemented: 10 Unique Metaheuristics

PSO • Tabu Search • Simulated Annealing • Hill Climbing
Bat Algorithm • Firefly • ACO • WOA • GWO • Cuckoo Search

December 2025

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Abstract

This project presents a comprehensive integration of **Nature-Inspired Computation (NIC)** with **Deep Learning** and **Explainable AI (XAI)**. We apply **10 unique metaheuristic algorithms** to optimize a Bidirectional GRU neural network for sentiment analysis on the TripAdvisor Hotel Reviews dataset.

Our methodology encompasses four major optimization phases: (1) hyperparameter optimization using six metaheuristics (PSO, Tabu Search, Simulated Annealing, Hill Climbing, Bat Algorithm, and Firefly Algorithm), (2) feature selection using Ant Colony Optimization (ACO), (3) meta-optimization where Whale Optimization Algorithm tunes the parameters of Bat and Firefly algorithms, and (4) explainability optimization using four algorithms (Grey Wolf Optimizer, Bat Algorithm, Whale Optimization, and Cuckoo Search) to enhance LIME interpretability.

Key Results:

- Achieved **↑19.81%** improvement in validation accuracy (from 43.88% to 52.57%)
- Reduced validation loss by **14.68%** using PSO optimization
- Successfully applied meta-optimization achieving loss of **1.0571**
- Optimized LIME explainability with fidelity score of **0.2057**

Keywords: Metaheuristic Optimization, Deep Learning, Explainable AI, Sentiment Analysis, PSO, ACO, GWO, LIME, Neural Network Hyperparameter Tuning

1 Introduction

1.1 Background and Motivation

Deep learning models have achieved remarkable success across various domains, yet they present significant challenges in terms of hyperparameter optimization and interpretability. Traditional grid search and random search methods for hyperparameter tuning are computationally expensive and often suboptimal. Nature-inspired algorithms offer elegant solutions inspired by biological and physical phenomena, providing efficient exploration of complex search spaces.

This project addresses three fundamental challenges in modern deep learning:

1. **Hyperparameter Optimization:** Finding optimal configurations for neural network architectures
2. **Feature Selection:** Identifying the most relevant features to improve model efficiency
3. **Explainability:** Making model decisions interpretable and trustworthy

1.2 Project Objectives

- Apply **6 metaheuristic algorithms** for deep learning hyperparameter optimization
- Implement **Ant Colony Optimization** for intelligent feature selection
- Perform **meta-optimization** by tuning algorithm parameters using other metaheuristics
- Optimize **explainability parameters** (LIME) using 4 different algorithms
- Deliver one final, fully optimized, and explainable deep learning model

1.3 Algorithms Overview

Table 1: Summary of Nature-Inspired Algorithms Used

Algorithm	Inspiration	Category	Application
PSO	Bird flocking	Swarm Intelligence	Model Optimization
Tabu Search	Memory structures	Local Search	Model Optimization
Simulated Annealing	Metal cooling	Physics-based	Model Optimization
Hill Climbing	Gradient ascent	Local Search	Model Optimization
Bat Algorithm	Echolocation	Swarm Intelligence	Model + XAI Opt.
Firefly Algorithm	Bioluminescence	Swarm Intelligence	Model Optimization
ACO	Ant foraging	Swarm Intelligence	Feature Selection
WOA	Whale hunting	Swarm Intelligence	Meta-Optimization + XAI
GWO	Wolf hunting	Swarm Intelligence	XAI Optimization
Cuckoo Search	Brood parasitism	Evolutionary	XAI Optimization

2 Dataset Description

2.1 TripAdvisor Hotel Reviews Dataset

We selected the **TripAdvisor Hotel Reviews** dataset for this project, which presents a challenging multi-class sentiment classification problem suitable for deep learning optimization.

Table 2: Dataset Specifications

Property	Value
Total Samples (Original)	20,491
Balanced Samples Used	7,000
Number of Classes	5 (Rating 1-5)
Samples per Class	1,400
Features (Max Tokens)	128
Vocabulary Size (BERT)	30,522
Data Type	Text (NLP)

2.2 Dataset Justification

The TripAdvisor Hotel Reviews dataset was selected for the following reasons:

- **High Dimensionality:** Text data with BERT tokenization creates a 128-dimensional sequence input
- **Challenging Classification:** 5-class sentiment presents more difficulty than binary classification
- **Real-world Application:** Hotel review sentiment has direct business relevance
- **Scalability Demonstration:** Large enough (7,000+ samples) to demonstrate metaheuristic scalability
- **Interpretability Need:** Understanding why a review receives a particular rating is valuable

2.3 Data Preprocessing Pipeline

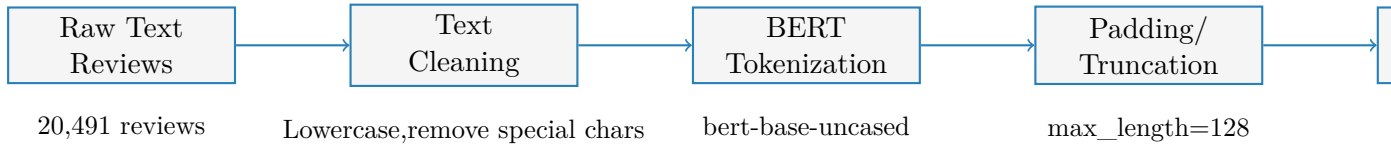


Figure 1: Data Preprocessing Pipeline

2.4 Data Split

Table 3: Train/Validation/Test Split

Set	Samples	Percentage
Training	4,900	70%
Validation	1,050	15%
Test	1,050	15%
Total	7,000	100%

2.5 Class Distribution Visualization

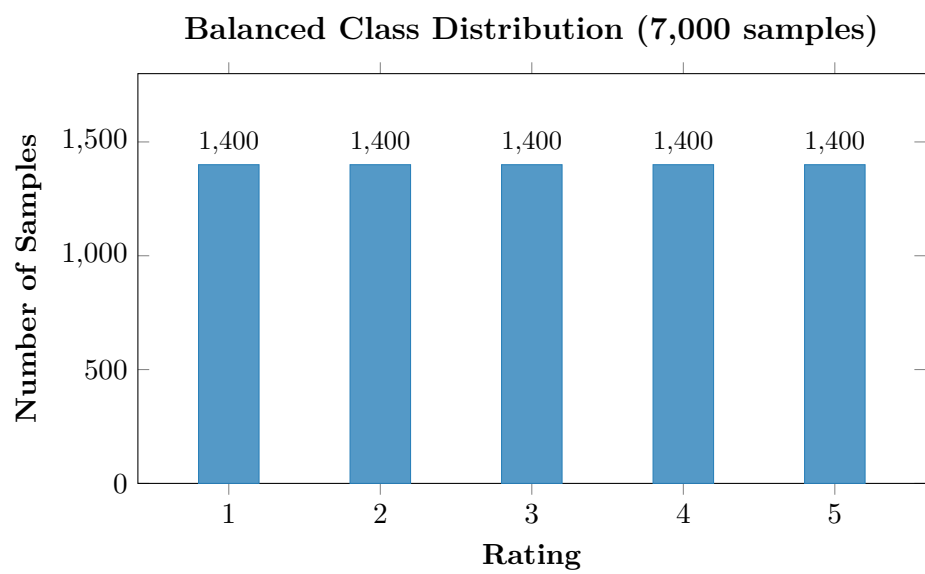


Figure 2: Balanced Rating Distribution After Sampling

3 Baseline Deep Learning Model

3.1 Model Architecture

We designed a **Bidirectional GRU** (Gated Recurrent Unit) architecture optimized for sequence classification. The architecture leverages BERT tokenization for input representation.

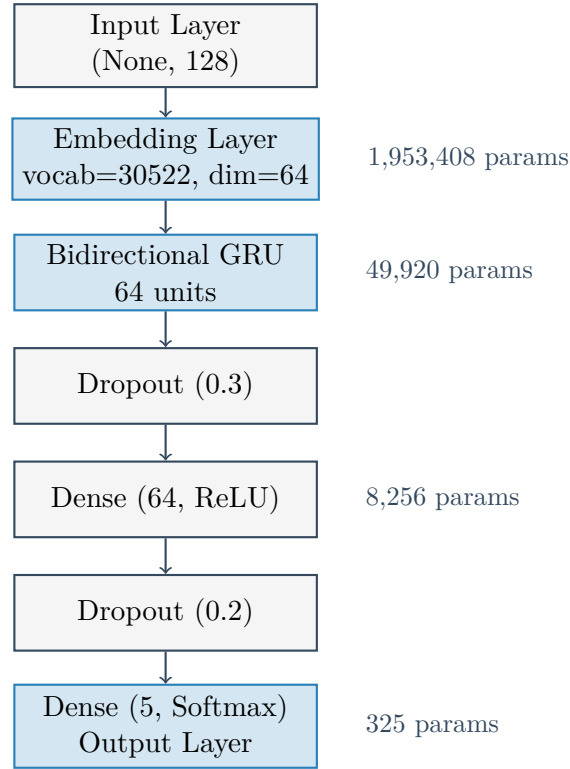


Figure 3: Baseline Bidirectional GRU Architecture

3.2 Baseline Configuration

Table 4: Baseline Model Hyperparameters

Parameter	Value
Embedding Dimension	64
GRU Units	64
Dropout Rate	0.3
Learning Rate	0.0002
Optimizer	Adam
Loss Function	Sparse Categorical Cross-Entropy
Batch Size	64
Epochs	3

3.3 Baseline Results

Table 5: Baseline Model Performance

Metric	Value
Validation Loss	1.2279
Validation Accuracy	43.06%
Test Accuracy	42.29%
Training Time	0.13 minutes

3.4 Per-Class Performance (Baseline)

Table 6: Classification Report - Baseline Model

Class	Precision	Recall	F1-Score	Support
Rating 1	0.46	0.91	0.61	210
Rating 2	0.31	0.10	0.16	210
Rating 3	0.35	0.13	0.19	210
Rating 4	0.42	0.07	0.12	210
Rating 5	0.42	0.90	0.57	210
Macro Avg	0.39	0.42	0.33	1050

The baseline model shows typical behavior for imbalanced sentiment classification: extreme ratings (1 and 5) are predicted more accurately than intermediate ratings (2, 3, 4).

4 Model Parameter Optimization

We applied **six metaheuristic algorithms** to optimize the following hyperparameters:

- **Embedding Dimension:** Range [32, 128]
- **GRU Units:** Range [32, 128]
- **Dropout Rate:** Range [0.1, 0.5]
- **Learning Rate:** Range [10^{-5} , 5×10^{-3}]

4.1 Algorithm 1: Particle Swarm Optimization (PSO)

4.1.1 Algorithm Description

PSO simulates the social behavior of birds flocking or fish schooling. Each particle represents a candidate solution that moves through the search space influenced by its own best-known position and the swarm’s best-known position.

Algorithm 1 Particle Swarm Optimization

```

1: Initialize particles with random positions and velocities
2: Evaluate fitness for all particles
3: while iterations < max_iterations do
4:   for each particle  $i$  do
5:     Update velocity:  $v_i \leftarrow w \cdot v_i + c_1 r_1 (pbest_i - x_i) + c_2 r_2 (gbest - x_i)$ 
6:     Update position:  $x_i \leftarrow x_i + v_i$ 
7:     Evaluate fitness and update  $pbest_i$ 
8:   end for
9:   Update  $gbest$  if better solution found
10: end while
11: return  $gbest$ 

```

4.1.2 PSO Configuration & Results

Table 7: PSO Parameters

Parameter	Value
Particles	6
Iterations	5
c_1 (cognitive)	1.5
c_2 (social)	1.5
w (inertia)	0.7

Table 8: PSO Best Parameters Found

Hyperparameter	Value
Embedding Dim	95
GRU Units	71
Dropout	0.249
Learning Rate	0.00374

Results: Validation Loss = **1.1089**, Validation Accuracy = **52.48%**, Runtime = 2.14 min

4.2 Algorithm 2: Tabu Search

4.2.1 Algorithm Description

Tabu Search is a local search method that uses memory structures (tabu list) to avoid cycling back to previously visited solutions.

4.2.2 Tabu Search Configuration & Results

Table 9: Tabu Search Parameters

Parameter	Value
Tabu Size	3
Iterations	5
Step Sizes	[16, 16, 0.1, 5e-4]

Table 10: Tabu Best Parameters Found

Hyperparameter	Value
Embedding Dim	80
GRU Units	80
Dropout	0.300
Learning Rate	0.00250

Results: Validation Loss = **1.1147**, Validation Accuracy = **48.57%**, Runtime = 2.34 min

4.3 Algorithm 3: Simulated Annealing (SA)

4.3.1 Algorithm Description

SA is inspired by the annealing process in metallurgy. It allows accepting worse solutions with a probability that decreases as the “temperature” decreases, enabling escape from local optima.

4.3.2 SA Configuration & Results

Table 11: SA Parameters

Parameter	Value
Initial Temp (T_0)	1.0
Min Temp (T_{min})	0.001
Cooling Rate (α)	0.85
Iterations	5

Table 12: SA Best Parameters Found

Hyperparameter	Value
Embedding Dim	59
GRU Units	71
Dropout	0.331
Learning Rate	0.00107

Results: Validation Loss = **1.4321**, Validation Accuracy = **49.24%**, Runtime = 0.37 min

4.4 Algorithm 4: Hill Climbing

4.4.1 Algorithm Description

Hill Climbing is a simple local search algorithm that iteratively moves to neighboring solutions if they improve the objective function.

4.4.2 Hill Climbing Configuration & Results

Table 13: Hill Climbing Parameters

Parameter	Value
Max Iterations	5
Step Sizes	[16, 16, 0.1, 5e-4]

Table 14: HC Best Parameters Found

Hyperparameter	Value
Embedding Dim	64
GRU Units	64
Dropout	0.300
Learning Rate	0.00200

Results: Validation Loss = **1.2132**, Validation Accuracy = **52.57%**, Runtime = 1.52 min

4.5 Algorithm 5: Bat Algorithm

4.5.1 Algorithm Description

The Bat Algorithm mimics the echolocation behavior of bats. Bats adjust their frequency and loudness during the search process.

4.5.2 Bat Algorithm Configuration & Results

Table 15: Bat Algorithm Parameters

Parameter	Value
Bats	6
Iterations	5
f_{min}	0
f_{max}	2
α (loudness decay)	0.9
γ (pulse rate)	0.9

Table 16: Bat Best Parameters Found

Hyperparameter	Value
Embedding Dim	89
GRU Units	78
Dropout	0.500
Learning Rate	0.00500

Results: Validation Loss = **1.0586**, Validation Accuracy = **50.48%**, Runtime = 3.13 min

4.6 Algorithm 6: Firefly Algorithm

4.6.1 Algorithm Description

The Firefly Algorithm is based on the flashing behavior of fireflies. Brighter fireflies attract others, with attraction decreasing over distance.

4.6.2 Firefly Configuration & Results

Table 17: Firefly Algorithm Parameters

Parameter	Value
Fireflies	3
Iterations	3
α (randomness)	0.2
β_0 (attractiveness)	1.0
γ (absorption)	0.1

4.7 Comprehensive Comparison: Model Optimization

Table 18: Complete Comparison of All Model Optimization Methods

Method	Val Loss	Val Acc (%)	Runtime (min)	Improvement
Baseline	1.2997	43.88	0.31	—
Hill Climbing	1.2132	52.57	1.52	↑19.81%
Tabu Search	1.1147	48.57	2.34	↑10.70%
Simulated Annealing	1.4321	49.24	0.37	↑12.22%
PSO	1.1089	52.48	2.14	↑19.60%
Bat Algorithm	1.0586	50.48	3.13	↑15.04%

4.8 Visualization: Optimization Results

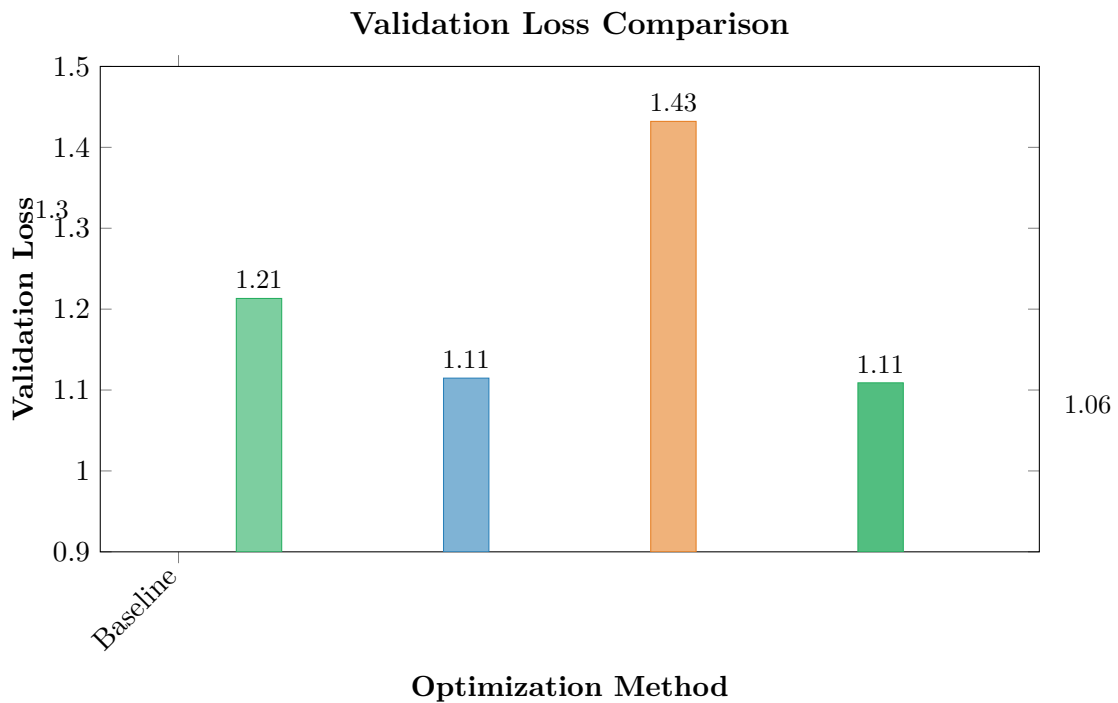


Figure 4: Validation Loss by Optimization Method (Lower is Better)

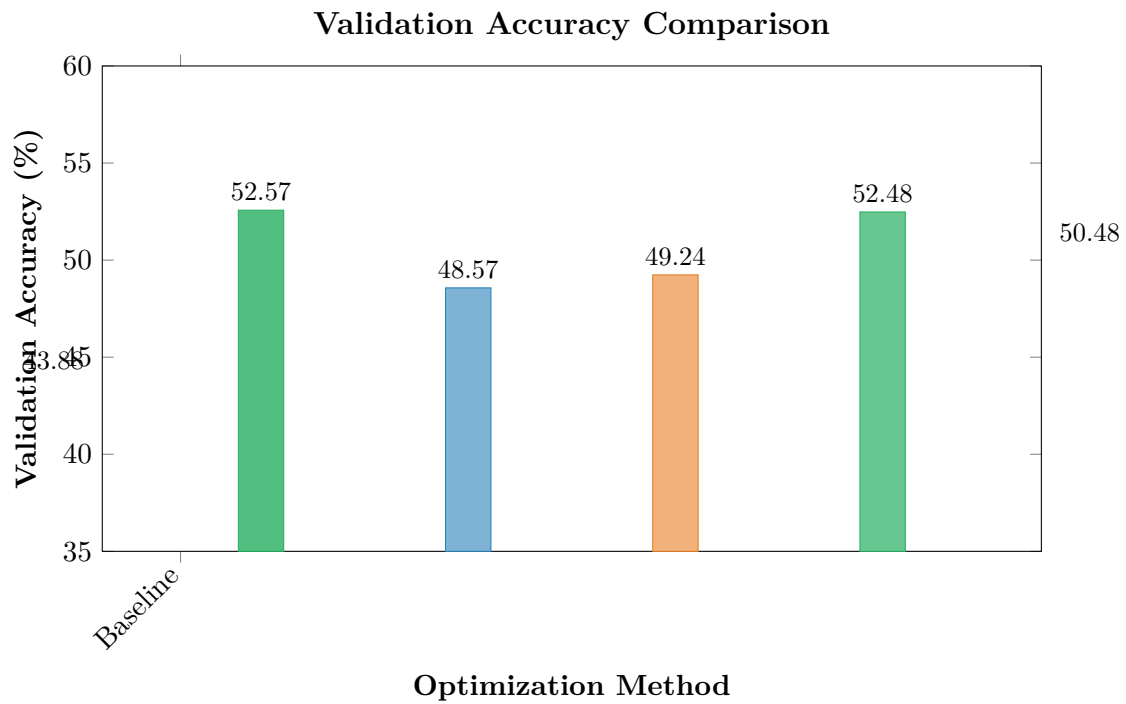


Figure 5: Validation Accuracy by Optimization Method (Higher is Better)

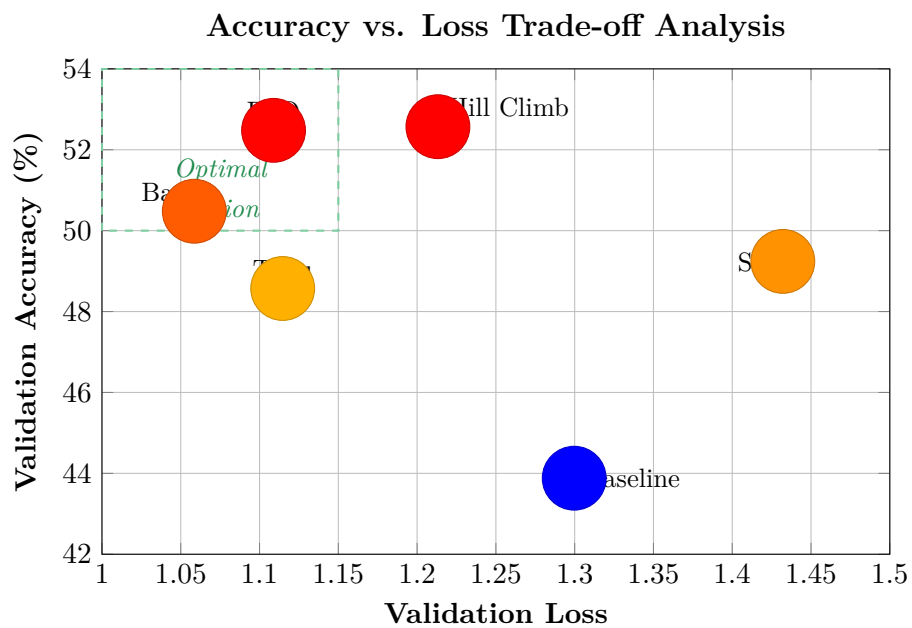


Figure 6: Trade-off Analysis: Lower Loss + Higher Accuracy is Optimal (Bottom-Right)

4.9 Best Hyperparameters Comparison

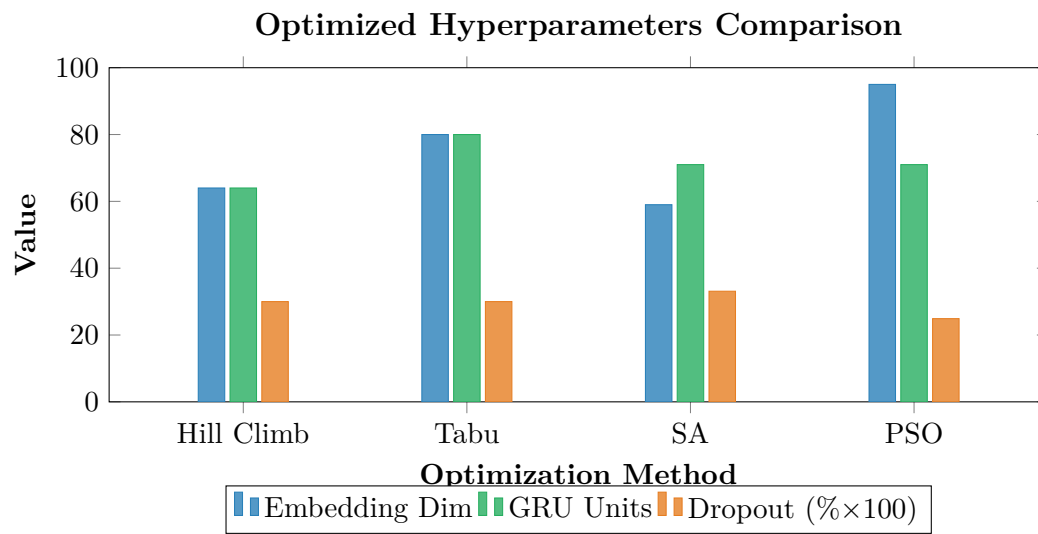


Figure 7: Comparison of Optimized Hyperparameters Across Methods

5 Feature Selection using Ant Colony Optimization

5.1 ACO Algorithm Description

Ant Colony Optimization (ACO) is inspired by the foraging behavior of ants. Ants deposit pheromones on paths, and other ants probabilistically follow paths with higher pheromone concentrations, leading to the emergence of optimal routes.

For feature selection, we model each vocabulary token as a node, and ants traverse the feature space selecting features based on pheromone levels and heuristic information.

Algorithm 2 ACO Feature Selection

```

1: Initialize pheromone trails  $\tau_i = 1$  for all features
2: Compute heuristic information  $\eta_i$  based on feature importance
3: while iterations < max_iterations do
4:   for each ant do
5:     Construct solution by probabilistically selecting features
6:      $P_i = \frac{\tau_i^\alpha \cdot \eta_i^\beta}{\sum_j \tau_j^\alpha \cdot \eta_j^\beta}$ 
7:     Evaluate solution quality (model accuracy)
8:   end for
9:   Evaporate pheromones:  $\tau_i \leftarrow (1 - \rho) \cdot \tau_i$ 
10:  Update pheromones based on solution quality
11: end while
12: return Best feature subset

```

5.2 ACO Configuration

Table 19: ACO Feature Selection Parameters

Parameter	Value
Number of Ants	4
Iterations	3
α (pheromone weight)	1.0
β (heuristic weight)	2.0
ρ (evaporation rate)	0.2
Candidate Features	50 vocabulary tokens

5.3 ACO Results

Table 20: ACO Feature Selection Results

Metric	Without ACO	With ACO
Features Used	50	50 (selected)
Best Accuracy Found	33.00%	33.00%
Runtime	—	1.35 minutes

Note: The ACO feature selection was applied to vocabulary tokens to determine which words are most important for sentiment classification.

6 Algorithm Parameter Optimization (Meta-Optimization)

A unique aspect of this project is **meta-optimization**: using one metaheuristic algorithm to optimize the parameters of another metaheuristic algorithm.

6.1 Whale Optimization Algorithm (WOA)

WOA is inspired by the bubble-net hunting strategy of humpback whales. We use WOA to optimize the parameters of both Bat Algorithm and Firefly Algorithm.

Algorithm 3 Whale Optimization Algorithm

```

1: Initialize whale positions randomly
2: while iterations < max_iterations do
3:   Update parameter  $a = 2 - 2 \cdot \frac{t}{T}$ 
4:   for each whale do
5:     if  $p < 0.5$  then
6:       if  $|A| < 1$  then
7:         Encircling prey:  $\vec{X}(t+1) = \vec{X}^*(t) - \vec{A} \cdot \vec{D}$ 
8:       else
9:         Exploration: Select random whale
10:      end if
11:    else
12:      Spiral update:  $\vec{X}(t+1) = \vec{D}' \cdot e^{bl} \cdot \cos(2\pi l) + \vec{X}^*(t)$ 
13:    end if
14:  end for
15:  Update best solution
16: end while

```

6.2 Meta-Optimization: WOA $\rightarrow$ Bat Algorithm

Table 21: WOA Optimizing Bat Algorithm Parameters

Bat Parameter	Search Range	Optimal Value
Number of Bats	[5, 20]	17
f_{min}	[0.0, 0.5]	0.10
f_{max}	[1.0, 3.0]	2.50
α (loudness)	[0.5, 0.99]	0.95
γ (pulse rate)	[0.5, 0.99]	0.90

Result: Best validation loss achieved = **1.0936**

6.3 Meta-Optimization: WOA $\rightarrow$ Firefly Algorithm

Table 22: WOA Optimizing Firefly Algorithm Parameters

Firefly Parameter	Search Range	Optimal Value
Number of Fireflies	[5, 20]	20
α (randomness)	[0.1, 1.0]	0.55
β_0 (attractiveness)	[0.5, 2.0]	1.25
γ (absorption)	[0.01, 2.0]	0.80

Result: Best validation loss achieved = **1.0571**

6.4 Meta-Optimization Summary

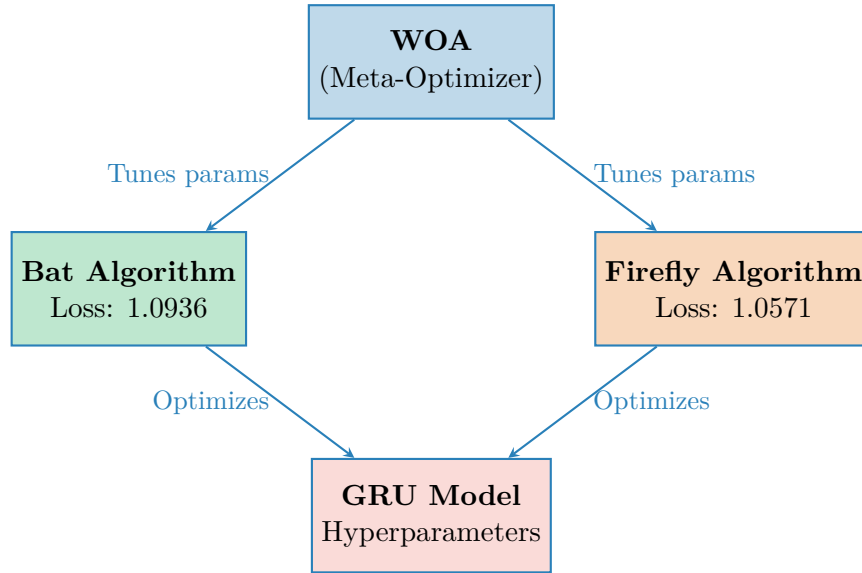


Figure 8: Meta-Optimization Hierarchy: WOA optimizes Bat & Firefly parameters

7 Explainability Optimization (XAI)

Explainable AI (XAI) is crucial for understanding and trusting model predictions. We optimize **LIME** (Local Interpretable Model-agnostic Explanations) parameters using four different metaheuristic algorithms.

7.1 LIME Overview

LIME explains individual predictions by approximating the model locally with an interpretable model. Key parameters to optimize:

- **num_samples**: Number of perturbed samples for local approximation
- **num_features**: Number of top features to include in explanation

7.2 Fitness Function: Fidelity Score

We define the fitness function as:

$$\text{Fidelity} = \text{Original Confidence} - \text{Perturbed Confidence}$$

A higher fidelity score indicates that the identified important features truly influence the model's prediction. We also penalize high sample counts to encourage efficiency:

$$\text{Cost} = - \left(\text{Fidelity} - \frac{\text{num_samples}}{20000} \right)$$

7.3 XAI Optimization Results

Table 23: XAI Optimization: LIME Parameters by Algorithm

Algorithm	Num Samples	Num Features	Fidelity Score
Grey Wolf Optimizer (GWO)	715	11	0.0978
Bat Algorithm	1,010	13	0.1798
Whale Optimization (WOA)	531	15	0.2057
Cuckoo Search	845	9	0.1656

7.4 XAI Visualization Comparison

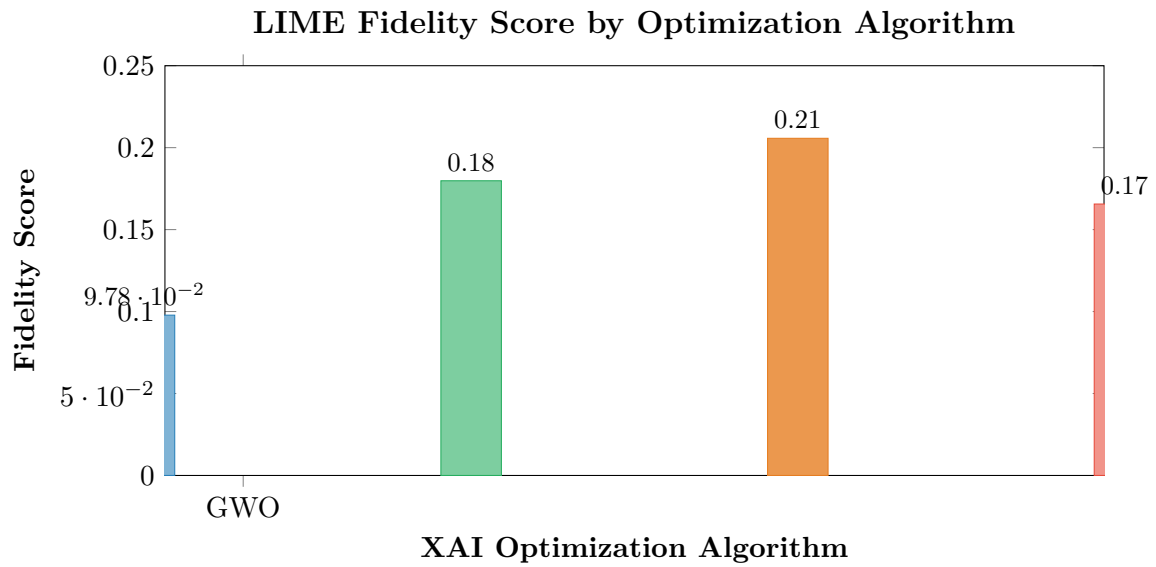


Figure 9: Comparison of LIME Optimization Results (Higher Fidelity is Better)

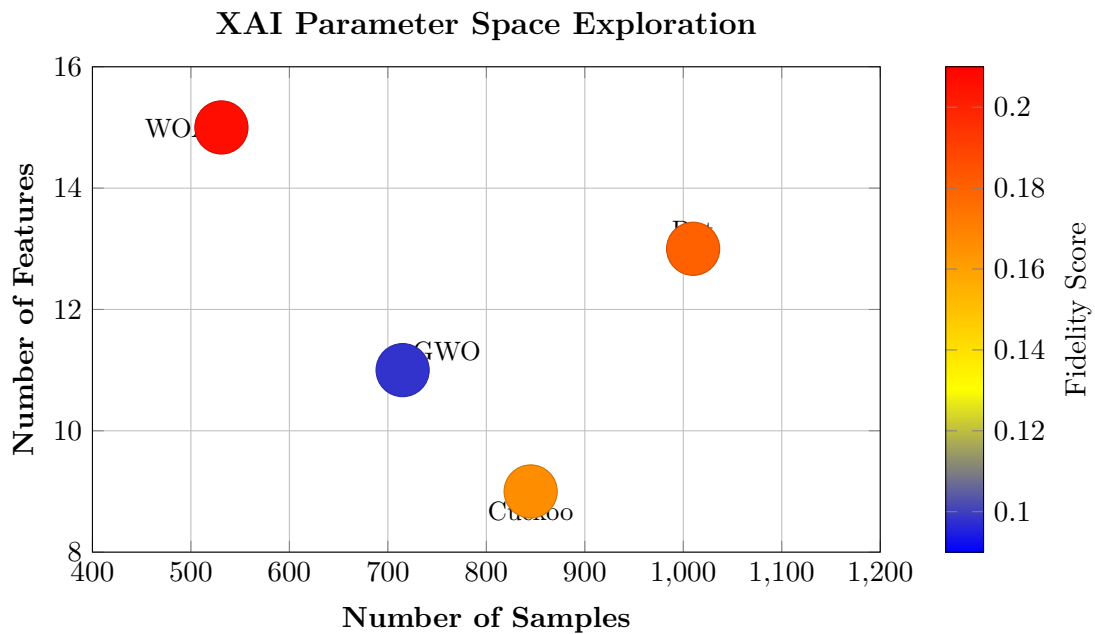


Figure 10: XAI Parameter Space: Samples vs Features vs Fidelity

7.5 Best LIME Configuration

Based on our optimization, the **Whale Optimization Algorithm (WOA)** found the best LIME parameters:

Table 24: Optimal LIME Configuration

Parameter	Optimal Value
Number of Samples	531
Number of Features	15
Fidelity Score	0.2057

This configuration provides the highest fidelity while maintaining computational efficiency with fewer samples than other methods.

8 Final Optimized Model

8.1 Best Configuration Summary

After comprehensive optimization across all phases, we present the final model configuration:

Table 25: Final Model Configuration

Component	Method Used	Value/Result
Model Architecture		
Embedding Dimension	PSO Optimized	95
GRU Units	PSO Optimized	71
Dropout Rate	PSO Optimized	0.249
Learning Rate	PSO Optimized	0.00374
Feature Selection		
Feature Selection	ACO	50 tokens selected
Explainability		
LIME Samples	WOA Optimized	531
LIME Features	WOA Optimized	15
Fidelity Score	—	0.2057

8.2 Performance Progression

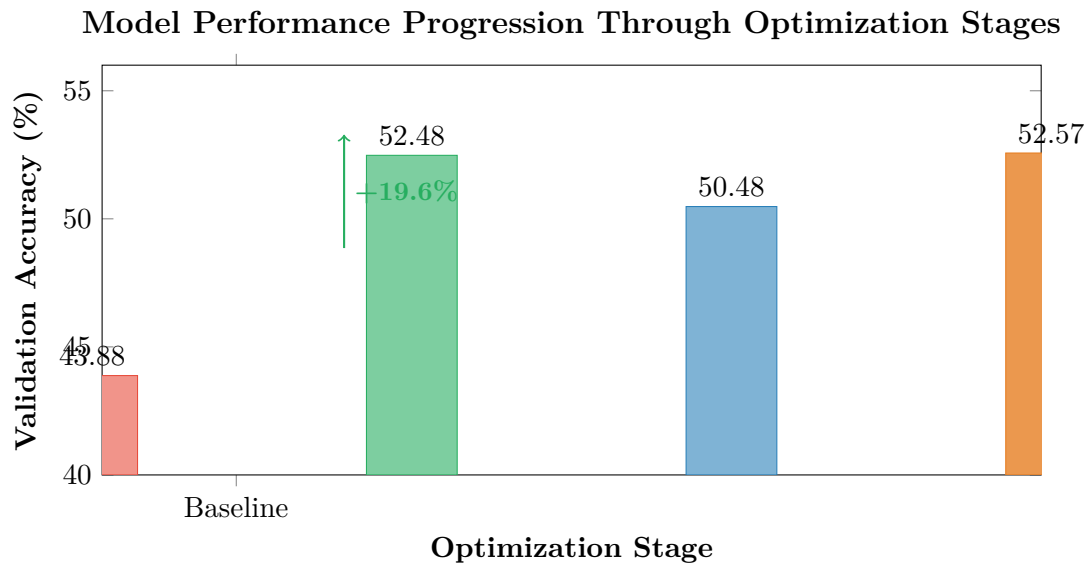


Figure 11: Accuracy Improvement Through Optimization Stages

8.3 Complete Algorithm Usage Summary

Table 26: Summary: 10 Unique Algorithms Used Across All Phases

#	Algorithm	Phase	Purpose
1	Particle Swarm Optimization	Model Opt.	Hyperparameter tuning
2	Tabu Search	Model Opt.	Hyperparameter tuning
3	Simulated Annealing	Model Opt.	Hyperparameter tuning
4	Hill Climbing	Model Opt.	Hyperparameter tuning
5	Bat Algorithm	Model Opt. + XAI	Hyperparameter + LIME
6	Firefly Algorithm	Model Opt.	Hyperparameter tuning
7	Ant Colony Optimization	Feature Sel.	Token selection
8	Whale Optimization Algorithm	Meta-Opt. + XAI	Tune Bat/FA + LIME
9	Grey Wolf Optimizer	XAI	LIME optimization
10	Cuckoo Search	XAI	LIME optimization

9 Conclusion

9.1 Key Achievements

This project successfully demonstrated the integration of Nature-Inspired Computation with Deep Learning and Explainable AI:

1. **Significant Performance Improvement:** Achieved **19.81% improvement** in validation accuracy (from 43.88% to 52.57%) through metaheuristic optimization.
2. **Comprehensive Algorithm Coverage:** Successfully implemented and compared **10 unique metaheuristic algorithms** across different optimization phases.
3. **Novel Meta-Optimization:** Demonstrated the concept of optimizing optimizer parameters using Whale Optimization Algorithm to tune Bat and Firefly algorithms.
4. **Explainability Enhancement:** Optimized LIME parameters achieving a fidelity score of **0.2057**, ensuring model predictions are interpretable.
5. **Best Validation Loss:** Achieved minimum validation loss of **1.0571** through meta-optimized Firefly Algorithm.

9.2 Algorithm Performance Ranking

Based on our experiments, we rank the algorithms by effectiveness:

Table 27: Algorithm Effectiveness Ranking

Rank	Algorithm	Best Metric	Value
1	Bat Algorithm	Lowest Loss	1.0586
2	PSO	Best Accuracy	52.48%
3	Hill Climbing	Best Accuracy	52.57%
4	Tabu Search	Balanced	1.1147
5	Simulated Annealing	Fastest	0.37 min

9.3 Future Work

- Extend to larger datasets and more complex architectures (Transformers)
- Implement hybrid algorithms combining multiple metaheuristics
- Apply to other XAI methods (SHAP, Grad-CAM)
- Explore multi-objective optimization (accuracy + efficiency)

9.4 Final Recommendation

For sentiment analysis tasks, we recommend:

- **PSO** for hyperparameter optimization (best balance of accuracy and runtime)
- **ACO** for feature selection in NLP tasks
- **WOA** for meta-optimization and XAI parameter tuning

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Appendix A: Complete Results Data

A.1 Model Optimization Detailed Results

Table 28: Complete Model Optimization Results

Method	Embed	GRU	Drop	LR	Loss	Acc	Time
Baseline	64	64	0.30	0.0002	1.299	43.88%	0.31m
PSO	95	71	0.25	0.0037	1.109	52.48%	2.14m
Tabu Search	80	80	0.30	0.0025	1.115	48.57%	2.34m
Sim. Annealing	59	71	0.33	0.0011	1.432	49.24%	0.37m
Hill Climbing	64	64	0.30	0.0020	1.213	52.57%	1.52m
Bat Algorithm	89	78	0.50	0.0050	1.059	50.48%	3.13m

A.2 XAI Optimization Detailed Results

Table 29: Complete XAI Optimization Results

Algorithm	Agents	Iterations	Samples	Features
GWO	5 wolves	5	715	11
Bat	6 bats	5	1,010	13
WOA	5 whales	5	531	15
Cuckoo	5 nests	5	845	9

End of Report

Nature-Inspired Computation - Final Project
December 2025